Loops and trace tables

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

TEACHER GUIDE · 1 OF 9

Prepare the purpose, prerequisites and materials

> Prepare Loops and trace tables: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Trace variable values before and after each **loop iteration**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-2/4, ai-2/5, python-bridge/6. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Branching and boundary tests

Without opening the old lesson, explain one key idea from Branching and boundary tests. Give an example and a mistake that the idea helps you avoid.

Look for: Trace the selected branch in an if elif else chain

Reference example, not the only acceptable wording: if tests negative first; elif checks <= 12 only after that fails; else handles the rest.

Unindented print runs after any selected branch, whose assignment gives label a value.

Expressions and operator order

Without opening the old lesson, explain one key idea from Expressions and operator order. Give an example and a mistake that the idea helps you avoid.

Look for: Evaluate numeric expressions using Python’s grouping rules

Reference example, not the only acceptable wording: 3 * 8 + 2 or (3 * 8) + 2 gives 26. 3 * (8 + 2) gives 30 and adds two to each of three packs, not two once.

Teach for **loop**s with finite integer ranges before adding more **loop** forms. Keep range values, body executions and **output** events separate. Reference: https://docs.python.org/3/tutorial/controlflow.html#the-range-function.

The examples use a positive default step and do not require lists or while **loop**s.

Outcomes

Trace variable values before and after each **loop iteration**

Use the excluded stop value in range correctly

Explain **initialization** and test zero one and several **iteration**s

Preparation

Python 3 editor and **loop**s.py

Counters or numbered paper cards

Blank **trace table** with turn total before and total after columns

Prerequisite answer

It reads the old 2, adds 3 and stores 5. Assignment updates a value; it is not an algebraic equality.

End goal for the pictorial case

Trace **loop**s and **trace table**s and explain the changed program before running it.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

TEACHER GUIDE · 2 OF 9

Explain and model the mechanism

> Model Loops and trace tables using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Explain Loops and trace tables · 1

Imagine adding numbered counters: first 1, then 2, then 3. A **loop** can repeat the same addition with a different number each time. This week you will predict each change before running a short Python script.

Tracing repeated calculations prepares you to inspect the data **process**ing used in larger AI applications.

Explain Loops and trace tables · 2

In for turn in range(1, 4), Python supplies 1, 2 and 3 in order. The start is included and the stop is excluded. The indented body runs once for each supplied value. The colon and four-space indentation mark that body.

This lesson uses integer range values with the default positive step of 1.

The **output** is 6. Before the first **iteration**, total is 0. Each assignment reads the previous total and adds the current turn. After turn 3, the range is exhausted, so execution continues after the **loop**.

The table records state; it does not create extra instructions.

Term	Meaning
Loop	A block of instructions that can run repeatedly
Iteration	One execution of a loop body
Accumulator	A variable that keeps a running result
Initialization	Giving a variable its starting value
Trace table	A record of variable values as instructions execute

State after each addition

TURN	TOTAL BEFORE	TOTAL AFTER
1	0	1
2	1	3
3	3	6

Initialize total to 0 before the loop. The unindented print runs once afterward.

range(1, 3) supplies 1 and 2, giving total 3. Stop 4 includes 3. Exit: the reset bug agrees for one **iteration** because there are no earlier additions to lose. range(1, 4) should give 6; the faulty version gives 3.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Range expression	Values supplied	Number of iterations
range(1, 4)	1, 2, 3	3
range(1, 2)	1	1
range(1, 1)	No values	0
range(4)	0, 1, 2, 3	4

Model the reasoning aloud · 3

With one argument, range starts at zero. range(4) supplies four values but none is 4. If the worked script uses range(1, 1), the body never runs and the initialized total remains 0.

Do not rely on a variable first assigned inside a **loop**: a zero-**iteration** run might never assign it.

> **Replace range(1, 4) with range(1, 3) in the worked script. Predict the supplied values and final total. What stop value is needed to include 3?**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Loops and trace tables

Accumulate a short range. A running total adds the integers visited by range(1,5).

The facts we will use

> **Supplied facts**
>
> Python 3: total=0; for value in range(1,5): total+=value; print(total). The print is after the **loop**.

AI ACADEMY / PYTHON-BRIDGE/6

Running total after each visited value

	Current value	Running total
Visit 1	1	1
Visit 2	2	3
Visit 3	3	6
Visit 4	4	10

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / PYTHON-BRIDGE/6

Loops and trace tables

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Visit the range

Values are 1,2,3,4; the stop value 5 is excluded.

2. Trace running total

Totals after each addition are 1,3,6,10.

3. Print after the loop

The final displayed value is 10.

CHANGE ONE THING

Replace range(1,5) with range(1,1).

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Visit the range	Values are 1,2,3,4; the stop value 5 is excluded.
2. Trace running total	Totals after each addition are 1,3,6,10.
3. Print after the loop	The final displayed value is 10.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **loop** repeats an operation for each visited value. A **trace table** distinguishes the current value from the **accumulator**; the range stop is not included.

What this example does not establish

Keep the stated limit

This bounded **loop** does not train an AI model; other **loop** conditions may require separate termination checks.

Change one thing in the pictured case

Change one thing

Change: Replace range(1,5) with range(1,1).

Predicted result

Prediction: No values are visited and the final **output** is 0.

Why the result follows

Reason: The **accumulator** starts at zero and an empty **loop** never changes it.

Ask while showing the picture

Ask while showing the picture · 4

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Loops and trace tables, reduce support deliberately and check individual reasoning before discussing answers.

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve assignment using the prerequisite.
7 to 20 minutes	Model range with numbered cards and complete W1.
20 to 33 minutes	Trace each row of the worked addition and predict output.
33 to 45 minutes	Complete W2 and compare predicted and observed values.
Session 2 start to 8 minutes	Retrieve included start and excluded stop.
8 to 20 minutes	Trace the reset bug, complete W3 and discuss the guided response.
20 to 35 minutes	Collect independent W4 code and zero one and several iteration traces.
35 to 45 minutes	Use W5 and the exit to examine assumptions and revealing tests.

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Use counters and let the learner move a pointer through one instruction at a time. Offer a blank trace grid and permit oral explanation.

Extension: put print inside the correct **loop** and explain why zero rounds produces no **output** while an outside print still does.

Reduce help deliberately

Completion practice

Completion task: Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each **iteration**. Predict the single **output**, then run and compare.

Reasoning to listen for

Teacher check: Rows are 2 / 0 / 2, 3 / 2 / 5, 4 / 5 / 9. Final **output** 9. Keep the observed result distinct from the **prediction**.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Loop**s and **trace table**s. Proposed explanation: "The stop value is included". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

> **Expected repair**
>
> Expected correction: List the supplied values before tracing; range(1, 4) excludes 4.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

List the four visited values and their running totals. Explain why 5 is absent and what prints for range(1,1).

Reasoned teacher answer

The visits are 1,2,3,4 and totals are 1,3,6,10. The stop is exclusive. An empty range leaves total at its initial 0.

Changed-case reasoning: No values are visited and the final **output** is 0. The **accumulator** starts at zero and an empty **loop** never changes it.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the investigation and diagnose misconceptions

> Run the Loops and trace tables investigation and use the first unsupported step to choose a teaching response.

The stop value is included: List the supplied values before tracing; range(1, 4) excludes 4.

Initialization belongs inside every **iteration**: Resetting the **accumulator** loses earlier work.

An unindented print repeats with the body: Indentation determines which statements repeat.

Zero **iteration**s mean the entire program does nothing: **Initialization** and statements after the **loop** still execute.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Loops and trace tables questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Guided and exit questions

range(1, 3) supplies 1 and 2, giving total 3. Stop 4 includes 3. Exit: the reset bug agrees for one **iteration** because there are no earlier additions to lose. range(1, 4) should give 6; the faulty version gives 3.

W1 Explain the range

Workbook question · W1

List the values supplied by range(2, 5) and explain why 5 is absent. How many times does its body run?

Teacher reasoning · W1

2, 3 and 4; the stop value is excluded. The body runs three times.

W2 Trace before running

Workbook question · W2

Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each **iteration**. Predict the single **output**, then run and compare.

Teacher reasoning · W2

Rows are 2 / 0 / 2, 3 / 2 / 5, 4 / 5 / 9. Final **output** 9. Keep the observed result distinct from the **prediction**.

W3 Repair the reset

Workbook question · W3

For the faulty script with range(1, 4), record total after each addition. Explain the wrong final **output** and remove only the faulty reset.

Teacher reasoning · W3

Faulty after-addition values are 1, 2, 3 because each starts from zero. Remove the indented total = 0. Correct after-addition values are 1, 3, 6; final **output** 6.

W4 Build a repeated addition independently

Workbook question · W4

A box starts with 5 counters. Each round adds 2 counters. Use rounds = 3, an initialized total and for turn in range(rounds). Print the final total once after the **loop**.

Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

Teacher reasoning · W4

Initialize total = 5 before the **loop**; add 2 inside it. For three rounds, turn values 0, 1, 2 give totals 7, 9, 11. Final **output**s for 0, 1, 3 rounds are 5, 7, 11. Print is unindented.

W5 Check a contract limitation

Workbook question · W5

In your counter script, range(-2) performs no **iteration**s. Does returning the initial 5 make negative rounds valid? Explain what the script checks and what it assumes.

Teacher reasoning · W5

No. The written contract requires a nonnegative integer. The script assumes this and does not reject negative values. Empty range behavior does not validate the **input**; report the limitation or add an explicit check.

Independent assessment and fresh retry

Teacher key · Independent assessment and fresh retry

W4 must preserve the initial 5, update by 2, trace three **iteration**s and print once. Require **output**s 5, 7, 11 for zero one and three rounds.

After support, retry with an initial 4 and additions of 3 for zero one and two rounds: **output**s 4, 7, 10. Record assistance before judging independence.

Assess independent transfer

Independent question · withhold the key

Selected case: List every value visited and each running total. Then predict the result for

range(2,2).

```
total = 0
for value in range(2, 6):
    total = total + value
print(total)
```

Selected case answer

Teacher key · Selected case answer

Values 2,3,4,5 give totals 2,5,9,14. Range(2,2) is empty, so total remains 0. The upper endpoint is excluded.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

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Connect the project, support and next lesson

> Connect Loops and trace tables to Python readiness portfolio, adapt support and plan a later changed-case check.

Add a labelled **loop**s and **trace table**s **evidence** sheet to Python readiness portfolio. Show how you can trace variable values before and after each **loop iteration**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Use counters and let the learner move a pointer through one instruction at a time. Offer a blank trace grid and permit oral explanation.

Extension: put print inside the correct **loop** and explain why zero rounds produces no **output** while an outside print still does.

Week 7 stores several values in a list and distinguishes an index from the value at that position.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: python-bridge/7, python-bridge/9, python-bridge/12. Use actual classroom dates, adjusting for holidays and gaps.

One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Loops and trace tables lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Trace variable values before and after each **loop iteration**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each **iteration**. Predict the single **output**, then run and compare.

Supported practice answer

Reasoned practice answer: Rows are 2 / 0 / 2, 3 / 2 / 5, 4 / 5 / 9. Final **output** 9. Keep the observed result distinct from the **prediction**.

Check every learner before discussion: Use the worked case for **Loop**s and **trace table**s. Proposed explanation: “The stop value is included”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: List the supplied values before tracing; range(1, 4) excludes 4.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Replace range(1,5) with range(1,1). Require a **prediction** and a reason before comparing. Expected result: No values are visited and the final **output** is 0.

Reason: The **accumulator** starts at zero and an empty **loop** never changes it.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately

Criterion	Not yet	With support	Independent
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

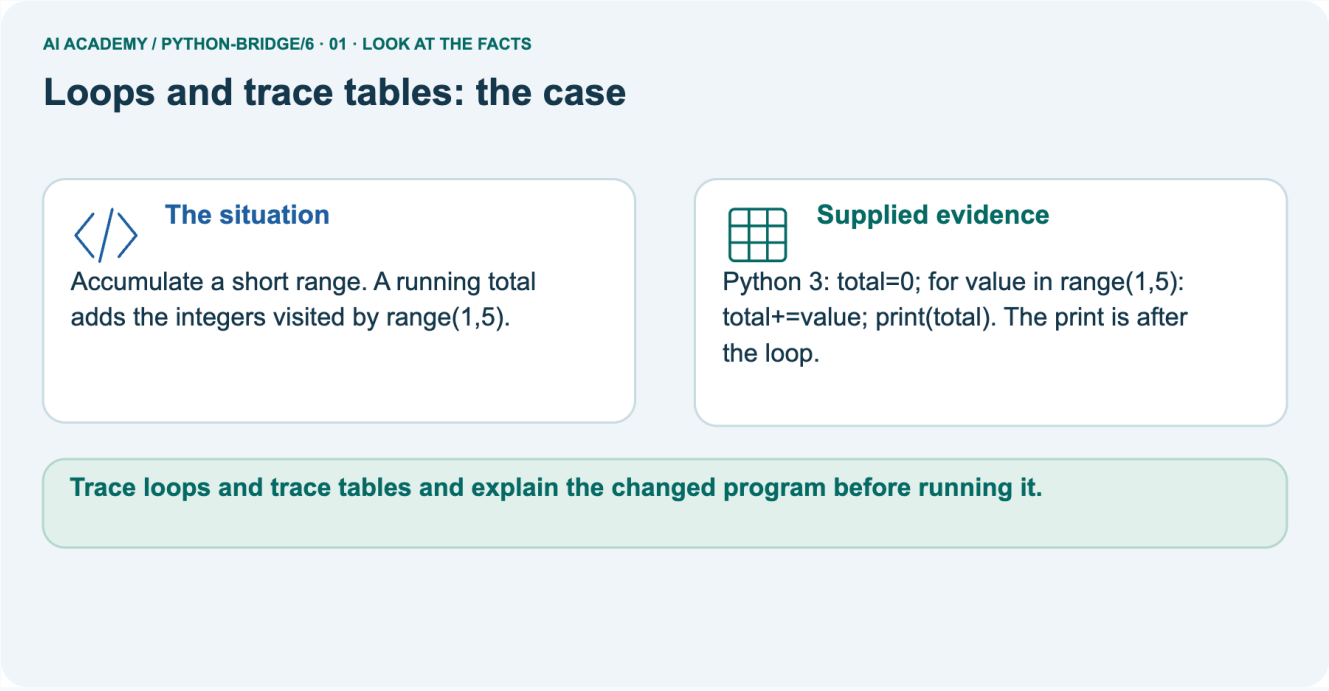
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Visual concept journey

> Show one Loops and trace tables graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Loops and trace tables: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

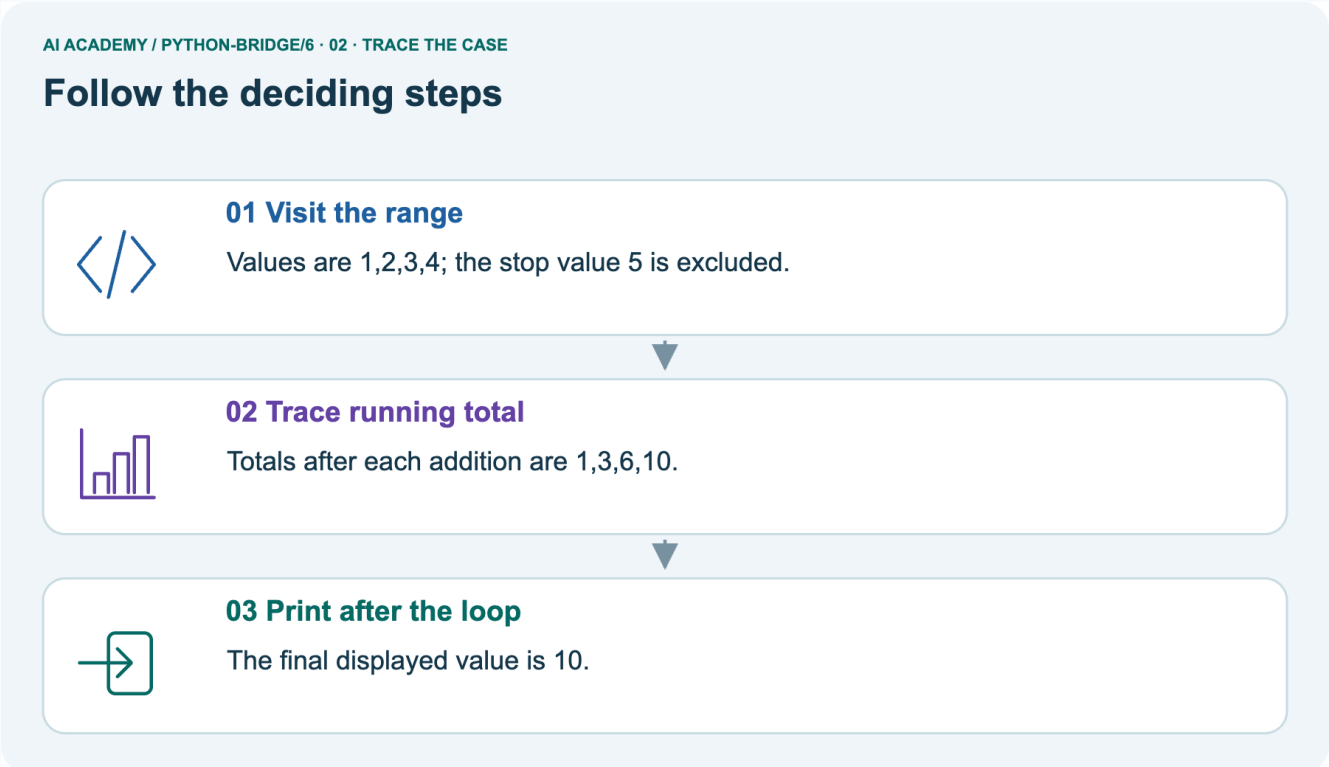
Accumulate a short range. A running total adds the integers visited by range(1,5). Supplied facts: Python 3: total=0; for value in range(1,5): total+=value; print(total). The print is after the **loop**.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Visit the range: Values are 1,2,3,4; the stop value 5 is excluded. → Trace running total: Totals after each addition are 1,3,6,10. → Print after the **loop**: The final displayed value is 10.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Running total after each visited value



Follow the labelled picture with the matching explanation. Open the vector diagram

Running total after each visited value. Visit 1: Current value = 1, Running total = 1. Visit 2: Current value = 2, Running total = 3. Visit 3: Current value = 3, Running total = 6.

Visit 4: Current value = 4, Running total = 10.

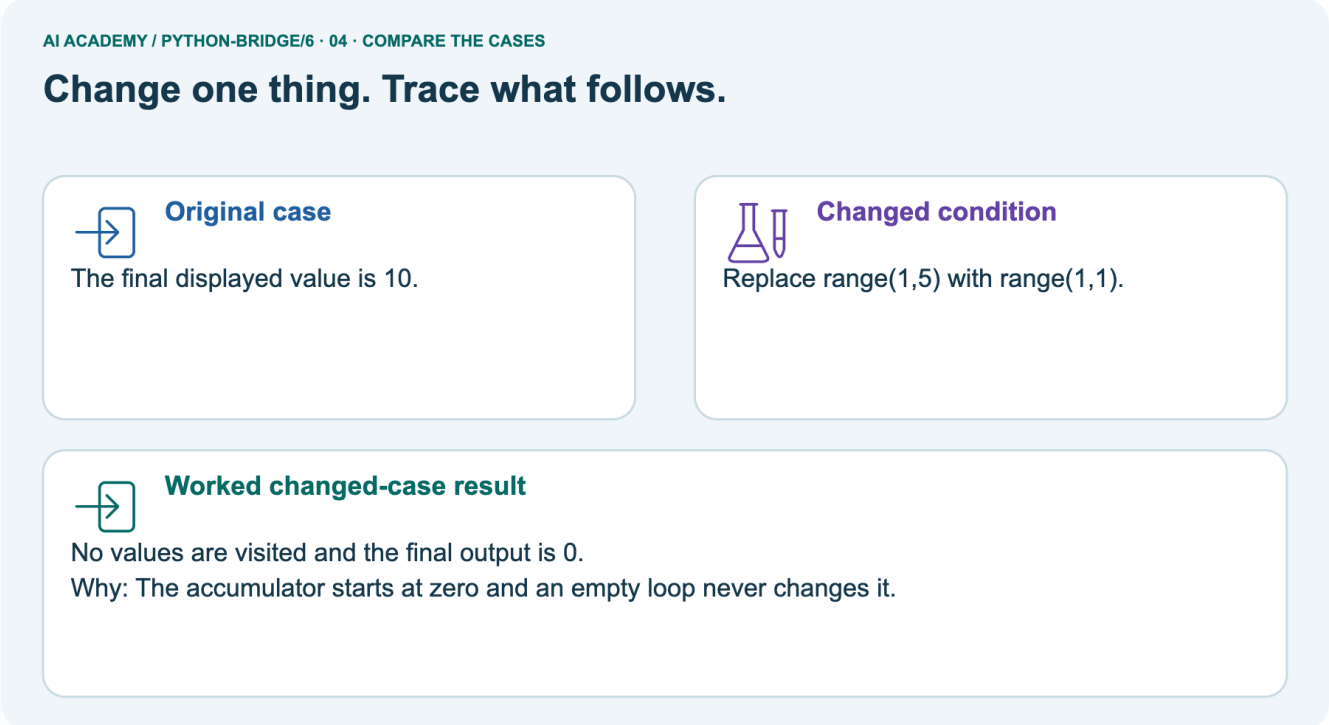
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The final displayed value is 10. Changed condition: Replace range(1,5) with range(1,1). Expected result: No values are visited and the final **output** is 0.

Reason: The **accumulator** starts at zero and an empty **loop** never changes it.

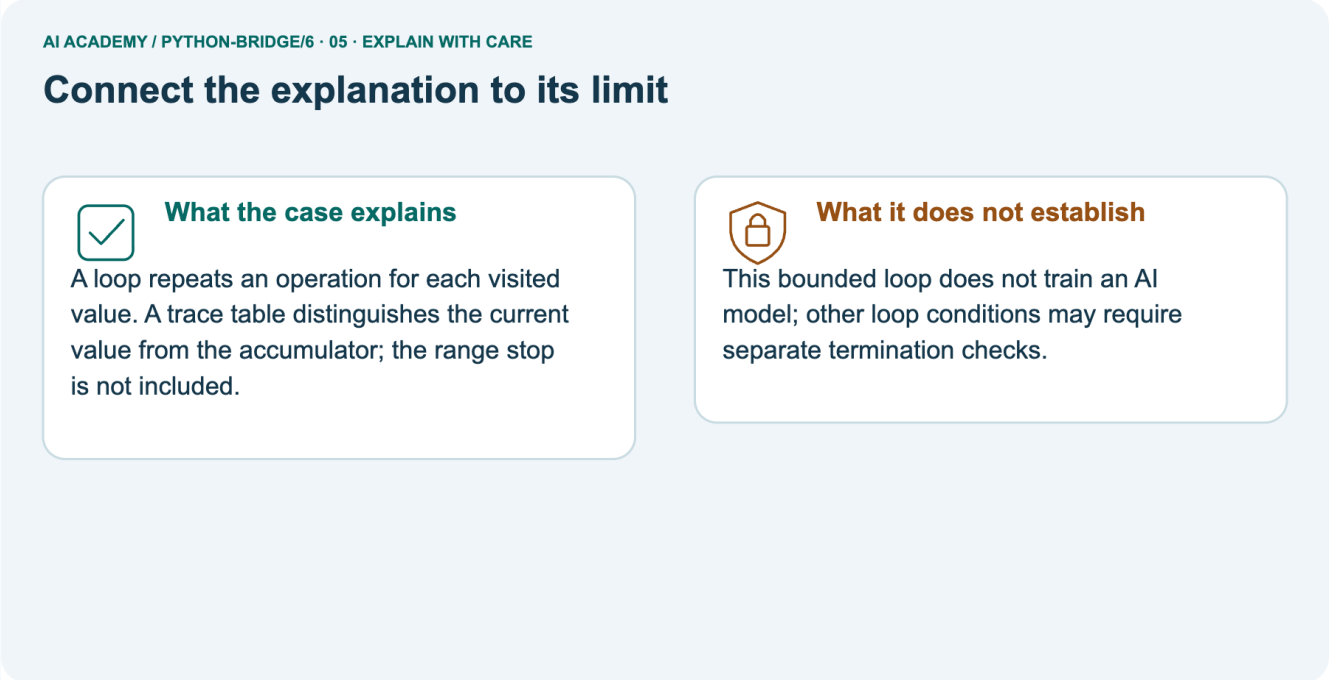
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The **accumulator** starts at zero and an empty **loop** never changes it.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **loop** repeats an operation for each visited value. A **trace table** distinguishes the current value from the **accumulator**; the range stop is not included.

Limit: This bounded **loop** does not train an AI model; other **loop** conditions may require separate termination checks.

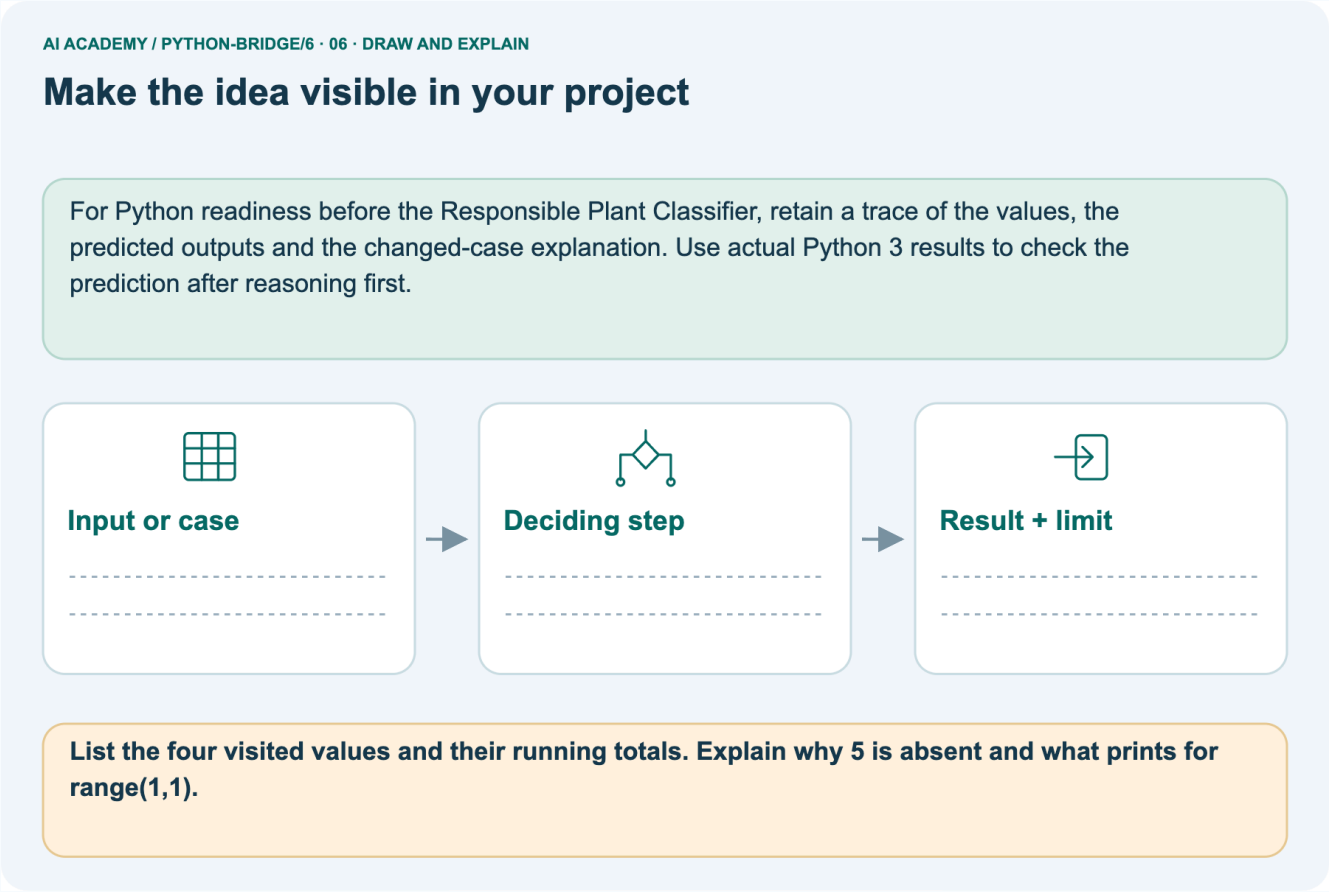
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This bounded **loop** does not train an AI model; other **loop** conditions may require separate termination checks.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: List the four visited values and their running totals. Explain why 5 is absent and what prints for range(1,1).

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The visits are 1,2,3,4 and totals are 1,3,6,10. The stop is exclusive. An empty range leaves total at its initial 0.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.